$p(73) = 1 - q(73)$

$= 1 - 0.1404 = 0.8596$

$p(74) = 1 - q(74)$

$= 1 - 0.2006 = 0.7994.$

Values of $L(x)$ are estimated using its relationship with $l(x)$ and $d(x)$ as follows:

$$L(72) = \frac{l(72) + l(73)}{2}$$

$$= \frac{4412 + 3724}{2}$$

$= 4{,}068$

$$L(73) = \frac{l(73) + l(74)}{2}$$

$$= \frac{3724 + 3201}{2}$$

$$L(74) = l(74) - \frac{d(74)}{2}$$

$$= 3201 - \frac{642}{2}$$

$= 2880.$

The value of $T(x)$ is estimated for $x = 72$ and 73 from the given value of $T(74)$ and the estimated values of $L(72)$ and $L(73)$ as

$$T(73) = L(73) + T(74)$$

$= 3463 + 26567 = 30{,}030.$

$$T(72) = L(72) + T(73)$$

$= 4068 + 30030 = 34{,}098.$

The life expectancy of the ***cohort*** at the age $x = 72$, 73 and 74 years is estimated using the relationship

$$e^{\circ}(x) = \frac{T(x)}{l(x)}$$

as

$$e^{\circ}(72) = \frac{34098}{4412} = 7.73$$

$$e^{\circ}(73) = \frac{30030}{3724} = 8.06$$

$$e^{\circ}(74) = \frac{26567}{3201} = 8.30$$

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